

JEE MAIN REVISION MASTER: SEQUENCES AND SERIES SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 32 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Arithmetic & Geometric Progressions (AP, GP, AGP)	24 PYQs	Progressions
High	AM-GM-HM Inequality & Telescoping Series	8 PYQs	Inequalities & Sigma

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Arithmetic & Geometric Progressions (AP, GP, AGP)	$S_n = \frac{n}{2}[2a + (n-1)d]$, $S_n = \frac{a(1 - r^n)}{1 - r}$, $S_{\infty} = \frac{a}{1 - r}$; ($ r < 1$), $T_n = S_n - S_{n-1}$
AM-GM-HM Inequality & Telescoping Series	$\frac{a+b}{2} \geq \sqrt{ab} \geq \frac{2ab}{a+b}$, $\sum n^2 = \frac{n(n+1)(2n+1)}{6}$, $\sum n^3 = [\sum n]^2$, $\frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Arithmetic & Geometric Progressions (AP, GP, AGP) (Very High Yield — 24 PYQs)

Core Principle: nth term and sum formulas for AP and GP, infinite geometric series convergence, and Arithmetico-Geometric series summation.

■■ **Caution / JEE Pitfall:** The formula $S_{\infty} = \frac{a}{1 - r}$ is valid ONLY when $|r| < 1$. If $|r| \geq 1$, the infinite series diverges.

■ **Speed Trick:** To choose 3 terms in AP: $a - d, a, a + d$ (sum eliminates d). In GP: $a/r, a, ar$ (product eliminates r).

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "The sum of the first ten terms of an A.P. is 160 and the sum of the first two terms of a G.P. is 8. If the first term of the A.P. is equal ..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "Let $729, 81, 9, 1, \dots$ be a sequence and P_n denote the product of the first n terms of this sequence. If $\sum_{n=1}^{40} (P_n)^{1/n} = \dots$ " [Ans: N/A]
- [2026 Evening Shift] Q#2 (MCQ): "Consider the quadratic equation $(n^2 - 2n + 2)x^2 - 3x + (n^2 - 2n + 2)^2 = 0, n \in \mathbb{R}$. Let α be the minimum value of the product of its ..." [Ans: C]

3.2 AM-GM-HM Inequality & Telescoping Series (High Yield — 8 PYQs)

Core Principle: Arithmetic, geometric, and harmonic mean inequalities for positive numbers, sum of first n natural powers, and partial fraction telescoping.

■ **Caution / JEE Pitfall:** $AM \geq GM$ equality holds IF AND ONLY IF all participating positive numbers are equal ($a_1 = a_2 = \dots = a_n$).

■ **Speed Trick:** Whenever asked for the minimum value of a sum of positive terms whose product is constant (e.g. $x + (1/x)$), use $AM \geq GM$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "Let $\alpha = 3+4+8+9+13+14+\dots$ upto 40 terms. If $(\tan \beta)^{\alpha/1020}$ is a root of the equation $x^2 + x - 2 = 0$, $\beta \in (0, \pi/2)$ " [Ans: N/A]
- **[2026 Morning Shift] Q#1 (MCQ):** "Let the sum of the first n terms of an A.P. be $3n^2 + 5n$. Then the sum of squares of the first 10 terms of the A.P. is:..." [Ans: C]
- **[2026 Evening Shift] Q#1 (MCQ):** "The sum $\frac{1}{31} + \frac{1}{3 \cdot 2^{31} + 3} + \frac{1}{3 \cdot 2^3 + 3^{31} + 3 + 5} + \dots$ up to 8 terms, is :..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

The sum of the first ten terms of an A.P. is 160 and the sum of the first two terms of a G.P. is 8. If the first term of the A.P. is equal to the common ratio of the G.P. and the first term of the G.P. is equal to common difference of the A.P., then the sum of all possible values of the first term of the G.P. is:

- (A) $(34/9)$ [Correct]
- (B) $(34/13)$
- (C) $(32/9)$
- (D) $(32/13)$

Correct Answer: A

Solution Analysis: Let the first term of the A.P. be a and common difference be d . Then, from the question: first term of the G.P. = d common ratio of the G.P. = a . Now use the given conditions one by one. 1. Sum of first 10 terms of the A.P. We know, $S_{10} = (10/2)[2a + (10-1)d]$ So, $160 = 5(2a + 9d)$ $2a + 9d = 32 \dots (1)$ 2. Sum of first 2 terms of the G.P. The G.P. has first term d and common ratio a . So its first two terms are: d, da Their sum is 8 : $d + da = 8$ $d(1+a) = 8 \dots (2)$ 3. Express d from equation (1) From $2a + 9d = 32$ we get $9d = 32 - 2a$ $d = (32 - 2a)/9$ Substitute this in equation (2): $\frac{32 - 2a}{9}(1+a) = 8$

Benchmark Problem 2: JEE Main 2026 (Online) 5th April Morning Shift (Q#1)

Let the sum of the first n terms of an A.P. be $3n^2 + 5n$. Then the sum of squares of the first 10 terms of the A.P. is:

- (A) 10220
- (B) 12860
- (C) 15220 [Correct]
- (D) 19780

Correct Answer: C

Solution Analysis: Given, $S_n = 3n^2 + 5n$ where S_n is the sum of the first n terms of the A.P. To get the n th term, we use $a_n = S_n - S_{n-1}$. So, $S_{n-1} = 3(n-1)^2 + 5(n-1) = 3(n^2 - 2n + 1) + 5n - 5 = 3n^2 - 6n + 3 + 5n - 5 = 3n^2 - n - 2$. Now, $a_n = (3n^2 + 5n) - (3n^2 - n - 2) = 6n + 2$. So the terms of the A.P. are: 8, 14, 20, ... Now we need the sum of squares of the first 10 terms: $\sum_{n=1}^{10} (6n+2)^2$. Expand: $(6n+2)^2 = 36n^2 + 24n + 4$. So, $\sum_{n=1}^{10} (6n+2)^2 = 36 \sum_{n=1}^{10} n^2 + 24 \sum_{n=1}^{10} n + 4 \sum_{n=1}^{10} 1 = 36 \cdot 11 + 24 \cdot 5 + 4 \cdot 10 = 396 + 120 + 40 = 556$. Now use the standard sums: $\sum_{n=1}^{10} n^2 = \frac{10 \cdot 11 \cdot 21}{6} = 385$.